

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP623
Subject Title	Optoelectronics: from Quantum Theory to Modern Technologies
Credit Value	3
Level	6
Pre-requisite/ Co-requisite/ Exclusion	NIL, but a general understanding of concepts in electromagnetism, quantum mechanics, optics, solid-state physics, and semiconductor physics is expected.
Objectives	This subject aims to introduce students to the basic quantum physics of the band theory and the interaction of photons with semiconductor materials. The course is also intended to present fundamental applications of optoelectronic devices to modern technologies such as solar cells, photodetectors, and light-emitting diodes.
Intended Learning Outcomes (Note 1)	Upon completion of the subject, students will be able to: (a) Understand the band theory of solids and the quantum theory of the fundamental electrical/optical properties of semiconductors. (b) Explain the working principles of modern optoelectronic technologies (c) Understand the high-density integration and industry-scale manufacture of optoelectronic devices (d) Experimentally carry out the micro-fabrication and characterization of solar cells
Subject Synopsis/ Indicative Syllabus (Note 2)	Fundamental quantum theory of semiconductors and optoelectronic processes. Photon-to-electron conversion devices such as photodetectors, solar cells, fundamental device limits, and large-scale integration. Basic optical properties such as light absorption, spontaneous and stimulated emission of photons in semiconductors, non-radiative transitions, and gain. Light-emitting diodes, solid-state lighting, semiconductor lasers, quantum wells, and quantum dot lasers.
Teaching/Learning Methodology (Note 3)	The methodology includes classroom teaching, assignments, presentations, and laboratory experiments. Basic theory and knowledge as well as reviews of the latest research progress will be systematically introduced in lectures. Classwork and assignments related to the content of lectures will be used to enhance students learning.

Assessment Methods in Alignment with Intended Learning Outcomes (Note 4)	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Continuous Assessment	40	✓	✓	✓	✓
	2. Project	60	✓	✓	✓	✓
	Total	100 %				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:					
	The continuous assessment includes assignments and laboratory reports. Assignments will strengthen students’ basic knowledge and analytical skill to solve problems related to concepts and applications of modern optoelectronic technologies. Laboratory report consolidate students’ experimental skills and knowledge learnt from lab work.					
	The final project will be assigned during the second half of the semester. Students will utilize the knowledge gained in classroom and acquired through literature reading to explain a problem at the forefront of optoelectronic research. Student performance will be assessed through a written report and final presentation.					
	Hence, the proposed assessment methods are necessary to assess the intended learning outcomes (i.e., items a, b, c, & d).					
Student Study Effort Expected	Class contact:					
	▪ Lecture/Seminar				21 Hrs.	
	Other student study effort:					
	▪ Laboratory				18 Hrs.	
	▪ Self-study				81 Hrs.	
	Total student study effort				120 Hrs.	
Reading List and References	1. Optoelectronics Photonics: Principles Practices, 2 nd Edition; Author: Safa O. Kasap, Publisher: Pearson Education; (2012)					
	2. The Physics of Solar Cells (Properties of Semiconductor Materials) 1st Edition; Author: Jenny A Nelson, Publisher: Imperial College Press; (2003)					
	3. Electronic and Optical Properties of Semiconductor Structures; 1st edition; Author: Jasprit Singh; Cambridge University Press; (2007)					
	4. Optoelectronics; Author: M. H. Shahine (Ed.), Publisher: IntechOpen; (2021).					

Note 1: Intended Learning Outcomes

Intended learning outcomes should state what students should be able to do or attain upon subject completion. Subject outcomes are expected to contribute to the attainment of the overall programme outcomes.

Note 2: Subject Synopsis/Indicative Syllabus

The syllabus should adequately address the intended learning outcomes. At the same time, overcrowding of the syllabus should be avoided.

Note 3: Teaching/Learning Methodology

This section should include a brief description of the teaching and learning methods to be employed to facilitate learning, and a justification of how the methods are aligned with the intended learning outcomes of the subject.

Note 4: Assessment Method

This section should include the assessment method(s) to be used and its relative weighting, and indicate which of the subject intended learning outcomes that each method is intended to assess. It should also provide a brief explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes.